14. Division : Part 2

❁ Word Problems.

◆ **Divide 56 notebooks equally among 7 students. How many will each get ?**

Each one gets 8.

```
     8
7 ) 56
   − 56
    00
```

Exercise

Solve the following problems.

1. If the total cost of three compass boxes is ₹90, what is the cost of one compass box ? $3\overline{)90}$

2. If four kilograms of wheat cost ₹92, what is the cost of one kilogram of wheat ? $4\overline{)92}$

3. If 31 litres of milk are poured into 4-litre vessels, how many such vessels will be completely filled and how much milk will there be in the partly filled vessel ? $4\overline{)31}$

4. If 49 seedlings are to be planted in a garden in rows of 7 saplings each, how many such rows will be made ? $7\overline{)49}$

5. If 40 children stand in 5 equal rows for a drill, how many children will there be in each row ? $5\overline{)40}$

6. There are 87 beads. How many necklaces of 9 beads each can be made and how many beads will be left over? $9\overline{)87}$

Dividing a three-digit number by a one-digit number

Madhu has three hundred-rupee notes, six ten-rupee notes and nine one-rupee coins totalling ₹369. How should she divide it equally between Neena, Beena and Nagesh ?

```
     123
3) 369
  - 3
   06
  - 6
   009
   - 9
     00
```

First let us divide the 3 notes of 100 rupees.
3 ÷ 3 = 1, or, 3 can be divided by 3 once. That means each person gets one note of 100 rupees.
6 ten-rupee notes have to be divided between three people.
6 ÷ 3 = 2, so each person will get 2 notes of 10 rupees or 20 rupees.
9 rupees have to be divided between three people.
9 ÷ 3 = 3. Each person will get 3 rupees.
Each person gets 100 rupees + 20 rupees + 3 rupees = 123 rupees.
This division is shown vertically alongside.
The quotient is 123, so each person will receive ₹123.

Exercise

Divide the following.

(1) 4) 484 (2) 3) 396 (3) 4) 448 (4) 2) 468

◆ **Now let us divide 4 notes of 100 rupees, 6 notes of 10 rupees and 5 coins of 1 rupee, totalling ₹465 between 5 people.**

```
     0
5) 465
  - 0
    4
```

In these 465 rupees, there are 4 notes of 100 rupees. 5 can only be taken zero times from 4. This means that no one can get a 100-rupee note. Let us put a 0 in the hundreds place of the quotient.

```
     09
5) 465
  - 0
    46
  - 45
    01
```

We change 4 notes of 100 rupees for 10-rupee notes. Those 40 notes and the 6 notes we already have make a total of 46 ten-rupee notes. Let us divide them among 5 people. 5 can be subtracted from 46 a maximum of 9 times. 9 times 5 is 45. So, 46 – 45 = 1, and 1 note of 10 rupees remains.

```
     093
5) 465
  - 0
    46
  - 45
    015
   - 15
     00
```

We change this 10-rupee note for ten 1-rupee coins and add the 5 coins that we already have, making a total of 15 one-rupee coins to divide among 5 people. Thrice 5 is 15. Therefore, we can subtract 5, 3 times.

15 – 15 = 0. The quotient is 93.

When ₹465 are divided among 5 people, each person gets ₹93.

```
    2
3) 629
  - 6↓
    02
```

◆ Find : 629 ÷ 3

Divide the hundreds, tens and units in that order by 3.

Twice 3 is 6, so 6 can be divided by 3. We write 2 in the hundreds place of the quotient. Now let's bring down 2 tens and divide.

```
    209
3) 629
  - 6↓
    02
  -  0
    029
   - 27
    002
```

3 can be taken zero times from 2. Therefore, let us write 0 in the tens place of the quotient. 2 – 0 = 2, so 2 tens remain.
20 units from 2 tens and the 9 units we already have make a total of 29 units to be divided by 3. 9 times 3 is 27.
29 – 27 = 2.

Therefore, the quotient is 209 and the remainder is 2.

Exercise

Divide.

(1) 4) 494 (2) 2) 815 (3) 3) 242 (4) 5) 455

(5) 6) 578 (6) 8) 945 (7) 7) 647 (8) 4) 908

```
    4
2) 800
  - 8
    00
```

◆ 800 ÷ 2 = How much?

The dividend is 800 and the divisor is 2.
Let us divide 8H by 2; four times 2 is 8, so we write 4 in the hundreds place of the quotient.
8 – 8 = 0. Zero hundreds remain.

```
    40
2) 800
  - 8
    00
  - 00
    000
```

Now let us divide zero tens by 2; zero divided by any number is zero. Let us write zero in the tens place of the quotient.

Similarly, zero units divided by 2 is 0. Therefore, let us write 0 in the units place of the quotient. The quotient is 400 and the remainder is zero.

```
    400
2) 800
  - 8
    00
  - 00
    000
  - 000
    000
```

If, while dividing 800 ÷ 2, we do not put zeroes in the tens and units places, the quotient could be written by mistake as 40, or even 4, instead of 400.

Remember: While dividing, in case a number is divided zero times, write the zero in the correct place in the quotient.

Exercise

Divide the following.

(1) 500 ÷ 5 (2) 900 ÷ 6 (3) 120 ÷ 4

Word Problems

- **If 148 marbles were distributed, giving 4 to each child, how many children were given marbles ?**

```
     037
4) 148
  - 0
    14
  - 12
    028
  -  28
     00
```

Exercise

Solve the problems given below.

(1) If 126 peppermint sweets are divided equally between 9 children, how many sweets will each child get?

9) 126

(2) In a field, if 987 saplings are planted in 7 rows with an equal number of seedlings in each row, how many seedlings are planted in each row?

7) 987

(3) If a hostel has 132 students with 3 students in one room, in how many rooms are the students staying ?

3) 132

(4) How many bouquets can be made from 340 flowers, with 8 flowers in each bouquet? How many flowers will be left over ?

8) 340

(5) If we put 6 biscuits in one packet, how many packets can be made from 600 biscuits?

6) 600